

Tanachai Vilainate

Data Analyst | Carbon Capture Research

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ABOUT ME

Environmental Engineering graduate with hands-on experience in data analysis, KPI tracking, and reporting across industrial research projects. Skilled in Excel, SQL, and Power BI, with a track record of turning operational data into actionable insights, with growing expertise in machine learning and AI-assisted analytics

SKILLS

- Excel (Power Query, Power Pivot, Advanced)
- SQL (Joins, Subqueries, CTE, SET Operations)
- Python (Pandas, NumPy, Scikit-learn, Matplotlib)
- Machine Learning (Regression, Classification, Model Evaluation)
- Power BI
- Financial Statement Reading
- LLM-assisted workflows (Claude, ChatGPT)

CERTIFICATES

- Microsoft Office by Mahidol
- Excel Skills for Business Advanced
- ESG 101 Sustainability
- SQL for Data Science
- Soft Skills Training for Employment
- Intro to Machine Learning

LANGUAGES

- Thai (Native) | English (TOEIC 700)

EXPERIENCES

Asian Institute of Technology, Bangchak, and Mitico

September 2025 - May 2026

Department of Environmental Engineering and Management *Carbon Capture*

- Served as the focal point between AIT, Mitico, and Bangchak coordinating technical updates, troubleshooting, and scheduling across 3 stakeholder organizations.
- Consolidated and analyzed operational data from multiple sources (process logs, sensor readings, and manual records) using Excel to track 4 KPIs across multi-run experiments.
- Detected Gen 2 sorbent inefficiency through CO₂ capture/recovery analysis, identified it as the root cause of operational issues, and drove cost savings by discontinuing its production.
- Built Power BI visualizations to surface sorbent-performance trends and efficiency insights that informed sorbent selection and reduced operational costs.
- Delivered monthly progress reports and presented findings to 3 stakeholder organizations to support technical and operational decision-making.

Royal Thai Army Chemical Department

September 2024 - January 2025

Department of Engineering Royal Science

- Conducted water-quality testing (hardness/softener, TSS, pH) and recorded results for quality control and reporting.
- Supported R&D activities, documenting experimental data, prepared official technical documentation and reports.

Mass Rapid Transit Authority of Thailand

April 2023 - June 2023

Internship

- Monitoring environmental issues arising from the construction of MRT station and collecting the field data.
- Preparing reports, verifying accuracy against the original EIA document, and presenting to the supervisor to identify problems and propose solutions.

PROJECTS

Pilot-Scale Carbon Capture Unit

Industrial Pilot Data

- Built an interactive Power BI dashboard consolidating CO₂ capture and recovery data across 5 experimental runs (Gen 1, Gen 2.2, Zeolite) with KPI cards, bar/scatter charts, and slicers for run-by-run comparison.
- Visualized capture-efficiency differences across configurations, allowing non-technical stakeholders to interpret results without reviewing raw data
- Authored supporting documentation (Executive Summary, Recommendations, Assumptions) translating technical findings into business-facing insights for decision-making.

Coffee Shop Daily Revenue Prediction

Kaggle datasets

- Performed end-to-end analysis (ETL, dashboard, reporting) on Kaggle coffee shop dataset using Claude Cowork (Excel, and Python).
- Built a multiple linear regression model (Python, pandas, scikit-learn) to predict daily revenue, achieving R² of 0.9490 with Random Forest Model.
- Performed feature selection and statistical significance testing to identify 3 key drivers
- Found 3 significant impact factors; each \$1 increase in Average Order Value contributes \$243 to daily revenue.

EDUCATION

Kasetsart University

Bangkok, Thailand April 2024

- Bachelor of Engineering in Environmental Engineering, 3.30 GPA (2nd class honours)